Lesson 1 — Extra Practice

Prime Building Blocks of Numbers

Part A — Prime Factorization of Larger Numbers

1. Use a **division ladder** to find the prime factorization of each number. Start at 2 and work up through the primes; do not move to the next prime until the current one no longer divides evenly. Write each answer in exponent form.
(a) 6 468 (b) 9 075 (c) 13 860 (d) 4 913

2. Use a **factor tree** for each number. Choose a first factor pair that you can see quickly — it does not have to involve a prime — then keep branching until every leaf is prime.
(a) 5 184
(b) 12 167
(Hint: it is not even, and the digit sum is not a multiple of 3*. Try the larger primes.)*

3. Express each number as a product of prime factors in exponent form. Use whichever organized method you prefer.
(a) 10 296 (b) 23 400 (c) 29 095 (d) 74 088

4. Draw **two different** factor trees for 1 176: start one with the pair 4×294 and the other with the pair 8×147.
(a) Write the prime factorization you get from each tree.
(b) **Explain** what this shows about a number's prime factorization, and why the starting pair you choose never changes the final answer.

Part B — Counting Factors from the Prime Factorization

5. For each number, list *all* of its factors, state how many there are, and write its prime factorization in exponent form. Then look for a connection between the exponents and the count.
(a) 24 (b) 36 (c) 100

6. Question 5 points to a rule. If a whole number n has prime factorization $n = 2^a \times 3^b \times 5^c$, then the number of factors of n is

$$(_____) \times (_____) \times (_____).$$

Explain why the rule works. *(Think about how a factor of n is built: you must decide how many 2s to use, then how many 3s, then how many 5s. How many choices do you have at each step?)*

7. Use the rule from Question 6 to state how many factors each number has. **Do not list them.** Find the prime factorization first.
(a) 720 (b) 1 176 (c) 9 261 (d) 16 000

8. A number has prime factorization $2^6 \times 3^4 \times 5^2 \times 11$. How many factors does it have? Do *not* multiply the number out.

9. Compare $512 = 2^9$ with $180 = 2^2 \times 3^2 \times 5$.
(a) Which number is larger?
(b) Which has more factors?
(c) **Explain** why the bigger number does not automatically have more factors.

Part C — Working Backwards from the Factor Count

10. In this question you will find the **smallest** whole number that has exactly 12 factors.
(a) Using the rule from Question 6 backwards, list every way of writing 12 as a product of whole numbers greater than 1 (for example 12 itself, and 6×2). Each factor in the product is one more than a prime's exponent, so each way gives a pattern of exponents.
(b) For each pattern, build the smallest number that fits it by giving the largest exponent to the smallest prime.
(c) State the smallest number overall with exactly 12 factors.
(d) **Explain** why the largest exponent should always go on the smallest prime.

11. A number with exactly 3 factors is unusual.
(a) Find every whole number less than 100 that has exactly 3 factors.
(b) **Explain**, using the counting rule, why such a number must be the *square of a prime*. Your explanation should use the fact that 3 is prime.

12. Can a whole number have an **odd** number of factors?
(a) Count the factors of each whole number from 1 to 20 and list the ones whose count is odd. What kind of numbers are they?
(b) **Explain** using factor *pairs*: why does pairing normally give an even count, and what is different about the numbers in part (a)?
(c) **Explain** the same result a second way, using the rule $(a+1)(b+1)(c+1)\ldots$

Part D — Testing a Number for Primality

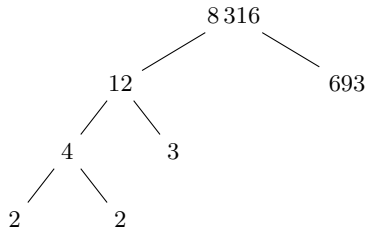
13. To test whether n is prime, you only need to divide by the primes up to $\sqrt{n}$.
(a) Suppose $n = a \times b$ where a and b are both whole numbers *larger* than $\sqrt{n}$. **Explain** why this is impossible, and what that tells you about where a factor of n must be found.
(b) **Explain** why you only need to test *prime* divisors, not every whole number below $\sqrt{n}$.

14. For each number, first state which primes you must test, then decide whether the number is prime or composite. If it is composite, give its prime factorization.
(a) 397 (b) 551 (c) 1 147 (d) 2 209

15. Kai tested 667 by dividing it by 2, 3, 5, 7, 11, 13, 17 and 19. None divided evenly, so he concluded that 667 is prime. **Explain** his error, then give the correct answer.

Part E — Error Analysis

16. A student drew this factor tree for 8 316 and wrote $8\,316 = 2^2 \times 3 \times 693$.



(a) Every step the student took is arithmetically correct, yet the answer is wrong. Identify the error and state the rule the student broke.
(b) Finish the tree and state the prime factorization of 8 316 in exponent form.
(c) How many factors does 8 316 have?

17. Another student argues: "1 divides into every number, so 1 is a prime factor of every number. That makes 1 prime."
(a) State the definition of a prime number and use it to show why 1 is not prime.
(b) **Explain** what would go wrong with prime factorizations if 1 were counted as prime. Use 12 as your example.
(c) **Explain** what would go wrong with the factor-counting rule from Question 6.

Part F — Stretch and Review Challenge

18. **Stretch.** Recall that $10! = 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 3\,628\,800$.
(a) Instead of factoring 3 628 800 directly, break each of the ten numbers into primes and collect them. Write 10! in exponent form.
(b) How many factors does 10! have?
(c) 10! ends in two zeros. **Explain**, using the prime factorization, exactly why there are two and not more.

19. **Stretch.** Find the **largest prime factor** of 58 435. Show the order in which you removed the primes, and explain how you knew when to stop.

20. *(Multiple Choice)* Which number has the **greatest** number of factors?
A. $2^3 \times 3^3$ B. $2^5 \times 3$
C. $2 \times 3 \times 5 \times 7$ D. $2^2 \times 3^2 \times 5$

21. *(Numerical Response)* The number of factors of $2^5 \times 3^3 \times 5^2 \times 7$ is _____.

22. *(Numerical Response)* The smallest whole number with exactly 10 factors is _____.

Lesson 2 — Extra Practice

Prime Factorization at Work: GCF and LCM

Part A — GCF and LCM Straight from the Exponents

1. When a factorization is written in **exponent form**, the GCF is built by taking the **lower** exponent of each prime the numbers *all* share. A prime missing from even one factorization cannot appear in the GCF. Determine the GCF of each group, and give your answer both in exponent form and as a whole number.

(a) $2^3 \times 3^2 \times 5$, $2^2 \times 3^3 \times 5^2$, $2^4 \times 3 \times 5^3$

(b) $2^2 \times 3 \times 7^2$, $2^3 \times 3^2 \times 7$, $2 \times 3^3 \times 7^3$

(c) 168, 252, and 420 (factor these yourself first)

2. The LCM is built the other way: take the **higher** exponent of each prime that appears in *any* of the factorizations. Determine the LCM of the three groups from question 1.

(a) the group in 1(a)

(b) the group in 1(b)

(c) the group in 1(c)

3. Let $a = 2^4 \times 3^2 \times 5$ and $b = 2^2 \times 3^5 \times 7$. Work entirely from the exponents — do not multiply the numbers out until the last step.

(a) Find the GCF of a and b in exponent form, then as a whole number.

(b) Find the LCM of a and b in exponent form, then as a whole number.

(c) **Explain** why the prime 5 appears in the LCM but not in the GCF.

4. Factor each group into primes, write the factorizations in exponent form, and then state both the GCF and the LCM.

(a) 126, 198, and 270

(b) 240, 360, and 600

5. Two numbers are $m = 2^4 \times 3^p \times 5^q$ and $n = 2^r \times 3^2 \times 5^3$. Their GCF is $2^2 \times 3^2 \times 5$ and their LCM is $2^4 \times 3^5 \times 5^3$.
(a) Find p, q, and r.
(b) State m and n as whole numbers.
(c) **Explain** how the GCF told you r before the LCM did.

Part B — The Identity GCF × LCM = ab

6. For any two whole numbers a and b, $\text{GCF}(a, b) \times \text{LCM}(a, b) = a \times b$. Test it on each pair: find the GCF, find the LCM, multiply them, and compare with $a \times b$.
(a) $a = 84$, $b = 90$
(b) $a = 2^4 \times 3^2 \times 5$, $b = 2^2 \times 3^5 \times 7$ (you already have both from question 3)

7. Once you trust the identity, any *three* of the four quantities a, b, GCF, LCM give you the fourth. Use it — listing multiples is not necessary.
(a) Two numbers have a GCF of 12 and an LCM of 180. One of the numbers is 36. Find the other.
(b) The GCF of 48 and 180 is 12. Find their LCM.

(c) Two numbers have a product of 3 024 and a GCF of 12. Find their LCM.
(d) Two numbers have a GCF of 15 and an LCM of 90. One of them is 30. Find the other.

8. **Explain** why the identity has to be true. Pick any prime p and suppose it appears m times in a and n times in b. Say how many times p appears in the GCF, how many times in the LCM, and why those two counts must add to $m + n$.

9. Two more consequences to think through.
(a) Ravi claims two whole numbers can have a GCF of 8 and an LCM of 60. Show that this is impossible, and state the general rule his example breaks.
(b) Does the identity extend to three numbers — is $\text{GCF}(a, b, c) \times \text{LCM}(a, b, c) = abc$? Test it on 4, 6, and 10, and state your conclusion.

Part C — Which Tool: GCF or LCM?

10. A rectangular sheet of veneer measures 180 cm by 252 cm. It is to be cut into identical square tiles with no material left over.

(a) What is the largest possible side length of a tile?

(b) How many tiles does the sheet produce at that size?

(c) **Explain** why the GCF — and not the LCM — answers this question.

11. Three bus routes all leave the terminal together at 6:00 a.m. Route A departs every 24 minutes, Route B every 36 minutes, and Route C every 40 minutes.

(a) At what time do all three next leave together?

(b) How many departures has each route made by that moment (not counting 6:00 a.m.)?

(c) **Explain** why the LCM — and not the GCF — answers this question.

12. Decide for yourself which tool each situation calls for, say which one you chose and why, then answer.

(a) Two rolls of ribbon, 168 cm and 216 cm, are cut into equal-length pieces that are as long as possible, with none wasted. How long is each piece, and how many pieces are there altogether?

(b) Two runners start together at the start line of a track. One laps in 72 seconds, the other in 90 seconds. When do they next cross the line together, and how many laps has each run by then?

13. A teacher has 84 pencils, 126 erasers, and 210 stickers to divide into identical supply bags with nothing left over.

(a) What is the greatest number of bags possible?

(b) What goes into each bag?

(c) If she wanted 21 bags instead, would that work? Explain.

Part D — Find the Error

14. Asked for the GCF of $a = 2^3 \times 3^2 \times 5^2$ and $b = 2^2 \times 3^4 \times 5$, a student wrote:

$$\text{GCF} = 2^3 \times 3^4 \times 5^2 = 16\,200$$

(a) Identify the mistake in one sentence.

(b) Give the correct GCF, in exponent form and as a number.

(c) The student's answer is not meaningless — what *is* 16 200 for these two numbers?

15. Asked for the LCM of 18 and 24, a student wrote: "A multiple of both is just $18 \times 24 = 432$, so the LCM is 432."

(a) The student is right that 432 is a common multiple. Why is it not the *lowest* one?

(b) Repair the method using the identity from Part B, and find the correct LCM.

(c) For which pairs of numbers does the student's shortcut actually give the right answer?

Part E — Stretch

16. Find **all** pairs of whole numbers whose GCF is 6 and whose LCM is 180.
Hint: write the two numbers as $6x$ and $6y$. What must be true of x and y, and what must $x \times y$ equal? Justify that your list is complete.

17. Find the smallest whole number that is divisible by every one of $1, 2, 3, \ldots, 10$.

(a) Give the number in exponent form and as a whole number.

(b) **Explain** why you do not need to consider 6, 8, 9, or 10 separately.

(c) Extend the list to 1 through 12. What changes, and what does not?

18. *(Multiple Choice)* The greatest common factor of $2^5 \times 3^4 \times 7$ and $2^3 \times 3^6 \times 5$ is
A. $2^3 \times 3^4$ B. $2^5 \times 3^6$
C. $2^3 \times 3^4 \times 5 \times 7$ D. $2^8 \times 3^{10} \times 5 \times 7$

Lesson 3 — Extra Practice

Perfect Squares and Perfect Cubes

Part A — Perfect Squares and Perfect Cubes from the Exponents

1. A number is a perfect square exactly when **every exponent in its prime factorization is even**, and a perfect cube exactly when **every exponent is a multiple of three**. Decide whether each number is a perfect square, a perfect cube, both, or neither. State any roots in exponent form.

(a) $2^6 \times 3^4$ (b) $2^9 \times 5^3$ (c) $2^6 \times 3^{12}$ (d) $2^4 \times 3^3 \times 5^2$

Part B — Making Perfect Squares and Perfect Cubes

2. Find the **smallest whole number** each of these must be multiplied by to give a perfect square, and state the square root of the result. (Raise each odd exponent to the next even number — nothing more.)

(a) $2^3 \times 3^5 \times 7^2$ (b) 540 (c) 1 176

3. Now find the smallest whole number each must be multiplied by to give a perfect **cube**, and state the cube root of the result. Parts (a) and (b) are the same numbers as in question 2 — compare the two multipliers when you are done.

(a) $2^3 \times 3^5 \times 7^2$ (b) 540 (c) 2 205

4. Dividing works the same way — strip off just enough to leave the exponents where you need them.

(a) Find the smallest whole number 2 000 can be *divided* by to leave a perfect square, and give the square root.

(b) Find the smallest whole number 2 000 can be *divided* by to leave a perfect cube, and give the cube root.

(c) Find the smallest whole number 72 must be *multiplied* by to give a number that is *both* a perfect square and a perfect cube.

Part C — Reasoning with Exponents

5. Decide whether each statement is **always true** or **not always true**. Explain with exponents, or give a counterexample.

(a) The product of two perfect squares is a perfect square.

(b) The sum of two perfect squares is a perfect square.

(c) If n is a perfect cube, then $8n$ is a perfect cube.

(d) If n is a perfect square, then $2n$ is a perfect square.

6. The number 36 is a perfect square.

(a) Write 36 and 36^3 in exponent form, without multiplying 36^3 out.

(b) Show that 36^3 is both a perfect square and a perfect cube, and state both roots.

(c) **Explain** why the cube of *any* perfect square is both a perfect square and a perfect cube.

Part D — Multiple Choice and Numerical Response

7. *(Multiple Choice)* The smallest whole number that $2^5 \times 3^4 \times 5$ must be *divided* by to leave a perfect cube is

A. 10 B. 12 C. 60 D. 120

8. *(Numerical Response)* The smallest whole number that 3 528 must be multiplied by to give a perfect cube is _____.

(Record your answer as a three-digit number.)

Lesson 4 — Extra Practice

Rational Numbers, Irrational Numbers, and Decimal Patterns

Part A — The Denominator Test

1. A fraction **in lowest terms** terminates exactly when its reduced denominator is built from 2's and 5's alone. **Reduce first** — the test is about the *reduced* denominator, not the one you were handed. For each fraction: reduce it, factor the new denominator, then predict *terminates* or *repeats*. Do not divide yet.

(a) $\frac{7}{40}$ (b) $\frac{15}{24}$ (c) $\frac{13}{60}$

(d) $\frac{42}{70}$ (e) $\frac{9}{56}$ (f) $\frac{22}{165}$

2. Now **confirm** four of your predictions by dividing. Write each answer in exact decimal form, putting the bar over the repeating block **only**.

(a) $\frac{15}{24}$ (b) $\frac{13}{60}$ (c) $\frac{9}{56}$ (d) $\frac{22}{165}$

3. Each denominator below contains a prime other than 2 or 5, so *most* numerators give a repeating decimal. Find one whole-number numerator k (with $0 < k <$ the denominator) that makes the fraction **terminate**, and show the reduced fraction and its decimal.

(a) $\frac{k}{90}$ (b) $\frac{k}{70}$ (c) $\frac{k}{24}$ (d) $\frac{k}{45}$

4. **Explain.** In question 3 the numerator had to be a multiple of something in every case. State the general rule — given a denominator d, which numerators k make $\frac{k}{d}$ terminate? — and explain why that rule falls straight out of the reduce-first denominator test.

Part B — How Long Is the Repeating Block?

5. When you long-divide by n, the only remainders possible are $1, 2, \ldots, n-1$. As soon as a remainder repeats, the digits repeat — so the block can be **at most** $n-1$ **digits long**. For each fraction, state that maximum *before* dividing, then divide and record the block's actual length.

(a) $\frac{1}{9}$ max: _____

(b) $\frac{1}{11}$ max: _____

(c) $\frac{1}{7}$ max: _____

(d) $\frac{1}{13}$ max: _____

6. Write each of these as a repeating decimal, using $\frac{1}{7} = 0.\overline{142857}$ as a starting point.

(a) $\frac{2}{7}$

(b) $\frac{3}{7}$

(c) $\frac{4}{7}$

(d) **What do you notice** about the six digits in every one of these blocks? Use the pattern to write $\frac{5}{7}$ without dividing.

7. When the reduced denominator has *both* kinds of prime factor, the decimal has a non-repeating lead followed by a repeating block. Split the denominator as $d = (2^a 5^b) \times m$, where m has no factor of 2 or 5. The lead is the **larger of** a **and** b digits long, and the block comes from m. Predict both lengths, then divide and write the decimal with the bar in the right place.

(a) $\frac{7}{12}$ lead: _____ block: _____

(b) $\frac{11}{30}$ lead: _____ block: _____

(c) $\frac{5}{22}$ lead: _____ block: _____

(d) $\frac{9}{56}$ lead: _____ block: _____

Part C — Turning a Repeating Decimal Back into a Fraction

8. Use the algebraic method: let x be the decimal, multiply by the power of 10 that shifts the repeating block left of the point, subtract, and solve. Give each answer in **lowest terms**.

(a) $0.\overline{8}$

(b) $0.\overline{72}$

(c) $0.\overline{135}$

9. These have a **non-repeating lead** after the decimal point, so you need *two* multiples of x: one with the block left of the point, and one with the block immediately right of it. Subtract those two. Give each answer in lowest terms.
(a) $0.16\overline{3}$ (b) $0.4\overline{27}$ (c) $1.2\overline{45}$

10. Same method, watching the sign and the whole-number part. Give each answer as an improper fraction in lowest terms.
(a) $-3.1\overline{6}$ (b) $2.\overline{306}$

11. **Explain.** In question 9 every subtraction left you with $990x$ or $900x$ on the left side. Without doing any more examples, state what the multiplier will be for a decimal with p non-repeating digits after the point and a repeating block of q digits, and explain where the 9's and the 0's come from.

Part D — Rational or Irrational?

12. Classify each number as **rational** or **irrational**. Several of these are traps — simplify *first*, then decide. Give a one-line reason for each.
(a) $\sqrt{16}$ (b) $\sqrt{18}$ (c) $\dfrac{\pi}{\pi}$ (d) $\sqrt{2}\times\sqrt{8}$

(e) $\sqrt{\dfrac{9}{25}}$ (f) $-\sqrt{0.04}$ (g) $0.\overline{45}$ (h) $\dfrac{\pi}{2}$

(i) $0.121\,221\,222\,12\ldots$ (j) $\sqrt{3}+\sqrt{3}$

13. **Reasoning.** Decide whether each statement is *always*, *sometimes*, or *never* true. If it is *sometimes*, give one example where it holds and one where it fails.
(a) The **sum** of two irrational numbers is irrational. (b) The **product** of two irrational numbers is irrational.

14. **Reasoning.** The sum of a rational number and an irrational number is *always* irrational. Suppose r is rational, s is irrational, and $r + s$ somehow came out rational. Show that this would force s to be rational too — and therefore cannot happen. (Hint: $s = (r + s) - r$.)

15. Between any two different numbers there is always a rational number. Find one for each pair below, and show with decimals that it really does sit between them.
(a) Between $\sqrt{7}$ and $\sqrt{8}$. (b) Between π and $\sqrt{10}$.

Part E — Error Analysis and Challenge

16. **Find the error.** Amira divides $5 \div 17$ on her calculator and the display reads 0.2941176471. She writes: *"The digits never repeat on the screen, so $\frac{5}{17}$ is irrational."*
(a) What is wrong with her reasoning? Answer **without** doing any division.
(b) The repeating block of $\frac{5}{17}$ is 16 digits long. Explain why a 10-digit display could never have shown it, and why question 5's remainder argument guarantees a block exists.

17. **Find the error.** Devon divides and gets $\frac{7}{12} = 0.58333\ldots$, then writes the answer as $0.\overline{583}$.
(a) Write out the first eight decimal places of what Devon's notation actually means.
(b) Write $\frac{7}{12}$ with the bar in the correct place, and state the one-sentence rule for where the bar goes.

18. **Stretch.** Show that $0.\overline{9} = 1$ exactly — not "very close to 1."
(a) Run the algebraic method of question 8 on $x = 0.\overline{9}$ and report what it gives.
(b) Give a second argument using $\frac{1}{3} = 0.\overline{3}$ and multiplying both sides by 3.
(c) If $0.\overline{9}$ were *less* than 1, some number would have to fit strictly between them. Explain why no number can.

19. *(Multiple Choice)* Which fraction has the **longest** repeating block? A. $\frac{1}{6}$ B. $\frac{1}{11}$ C. $\frac{1}{13}$ D. $\frac{1}{22}$

20. *(Numerical Response)* When the repeating decimal $0.2\overline{63}$ is converted to a rational number in lowest terms $\frac{a}{b}$, the value of $b - a$ is _____. *(Record your answer in the numerical response boxes from left to right.)*

Lesson 5 — Extra Practice

Classifying the Real Number System

Part A — Simplify First, Then Classify

1. Each number below is written in a form that hides what it really is. **Simplify it first**, then list *every* set from $N, W, I, Q, \overline{Q}, R$ that it belongs to, ordered from the largest set to the smallest set.

(a) $\sqrt{49}$ (b) $\sqrt[3]{-8}$ (c) $\dfrac{12}{4}$

(d) $0.\overline{45}$ (e) $\sqrt{0.25}$ (f) $-\sqrt{9}$

2. Same instructions. These take one more step of simplifying.

(a) $\sqrt[3]{0.027}$ (b) $\dfrac{\sqrt{18}}{\sqrt{2}}$ (c) $\pi - \pi$

(d) $\sqrt[3]{-\dfrac{27}{64}}$ (e) $\sqrt{2} \times \sqrt{50}$ (f) $\sqrt{3} + \sqrt{3}$

3. A number belongs to several sets at once, but only one of them is the **smallest** set that still contains it. For each number, simplify and then name that single set — the set it belongs to *in the strictest sense*.

(a) $\sqrt{121}$ (b) $\sqrt[3]{-1\,000}$ (c) $0.\overline{3}$ (d) $\sqrt{5}$ (e) $\sqrt{16} - 4$ (f) $-\sqrt{0.16}$

Part B — Nesting: Why the Sets Sit Inside One Another

4. Every natural number is a rational number, but not every rational number is a natural number.

(a) **Explain** why *every* number in N must also be in Q. Your explanation should show how to write any natural number in the form $\dfrac{a}{b}$ with $a, b \in I$ and $b \neq 0$.

(b) **Explain** why the reverse fails, and give *two* rational numbers that are not natural — one that is negative, and one that is positive but not an integer.

5. The word *nested* means one set sits entirely inside the next.

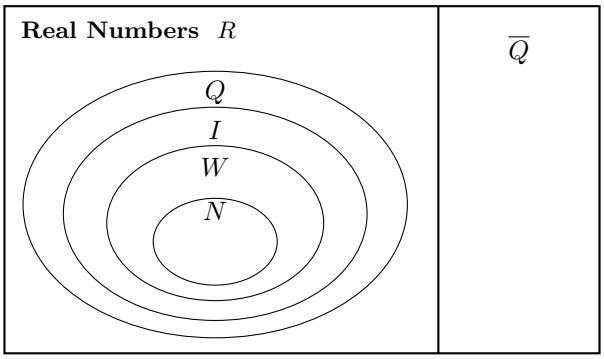
(a) Fill in the five set names in order, smallest to largest:
_____ inside _____ inside _____ inside _____ inside _____.

(b) **Explain** where $\overline{Q}$ belongs relative to that chain, and why it cannot be inserted anywhere *inside* it.

(c) $\sqrt{100}$ belongs to five sets. In the strictest sense it belongs to just one. Which one, and what does "strictest" mean here?

6. The diagram below shows the real number system: the rectangle is R, the vertical line separates the rationals from the irrationals, and the nested ellipses are Q, I, W, N. Write each of the eight numbers below into the **innermost** region of the diagram that it belongs in. Simplify first where you need to.

$$\sqrt{81}, \quad -\sqrt[3]{125}, \quad 0, \quad \frac{5}{8}, \quad -\pi, \quad \sqrt{0.09}, \quad \sqrt[3]{7}, \quad \frac{18}{3}$$



7. For each description, state whether **no number**, **exactly one number**, or **infinitely many numbers** fit it. Give an example whenever one exists.

(a) In W but not in N.

(b) In I but not in W.

(c) In Q but not in I.

(d) In I but not in Q.

(e) In R but in neither Q nor $\overline{Q}$.

(f) In $\overline{Q}$ but not in R.

Part C — Always, Sometimes, or Never

8. Complete each statement with *always*, *sometimes*, or *never*. Then **justify** it: if you answered *always* or *never*, say why it must be so; if you answered *sometimes*, give one example that works *and* one counterexample that does not.

(a) The sum of two integers is _____ a natural number.

(b) The quotient of two integers is _____ an integer.

(c) An irrational number times an irrational number is _____ irrational.

(d) The sum of a rational number and an irrational number is _____ irrational.

(e) A number whose decimal never terminates is _____ in the set $\overline{Q}$.

(f) A rational number times an irrational number is _____ irrational.

(g) A number in $\overline{Q}$ is _____ in the set R.

(h) The square root of a rational number is _____ rational.

Part D — Error Analysis

9. A student writes: "$\sqrt{16}$ has a radical sign, so it is irrational. It goes in $\overline{Q}$."

(a) Identify the mistake in one sentence.

(b) List every set $\sqrt{16}$ actually belongs to, largest to smallest.

(c) State a reliable rule for deciding when $\sqrt{n}$, $n \in N$, is irrational.

10. A second student writes: "0 is a natural number, because you can count to it."

(a) Identify the mistake, quoting the definition of N from this lesson.

(b) List every set 0 *does* belong to, largest to smallest, and state what makes 0 the one number that separates W from N.

11. A third student writes: "$0.\overline{45}$ is irrational, because its decimal goes on forever."
(a) Identify the mistake. Which *two* conditions must a decimal meet before it is irrational?

(b) Convert $0.\overline{45}$ to a fraction in lowest terms to settle the argument for good.

Part E — Closure

A set is **closed** under an operation if applying that operation to any two members of the set always produces a result that is *still in the set*. One counterexample is enough to show a set is not closed.

12. Answer each with yes or no, and support your answer.
(a) Is N closed under subtraction? If not, give a counterexample and name the smallest set that *is* closed under subtraction.

(b) Is I closed under division? If not, give a counterexample and name the smallest set that *is* closed under division of non-zero numbers.

13. Closed or not closed? Give a counterexample for every "not closed."
(a) N under addition

(b) W under subtraction

(c) I under multiplication

(d) Q under division

(e) R under subtraction

(f) W under division

14. The irrationals behave worse than any other set here.
(a) Is $\overline{Q}$ closed under addition? Give a counterexample.

(b) Is $\overline{Q}$ closed under multiplication? Give a counterexample.

(c) **Explain** in one or two sentences why $\overline{Q}$ fails where Q succeeds.

Part F — Stretch, Multiple Choice and Numerical Response

15. **Stretch.**
(a) Find a number that is in Q but not in I.
(b) Find a number that is in I but not in W.
(c) Find a number that is in R but not in Q.
(d) For each of (a)–(c), say how you know your number fails the smaller set.
(e) **Explain** why there is no number at all that is in Q and in $\overline{Q}$ at the same time. Base your explanation on the two definitions, not on examples — then say what Q and $\overline{Q}$ together make up.

16. *(Multiple Choice)* How many of the numbers

$$\sqrt{64}, \quad \sqrt[3]{-27}, \quad \frac{21}{7}, \quad 0.\overline{6}, \quad \sqrt{2.25}, \quad \frac{\sqrt{48}}{\sqrt{3}}$$

belong to the set I?
A. 3 **B.** 4 **C.** 5 **D.** 6

17. *(MC)* Consider these four statements.
i) Every number in W is also in Q.
ii) Every number in Q is also in I.
iii) N is closed under multiplication.
iv) $\overline{Q}$ is closed under division.
Which are **false**?
A. i) and iii) only **B.** ii) and iv) only
C. ii) only **D.** iv) only

18. *(Numerical Response)* Of the twelve numbers below, the number of them that belong to the set Q is _____. (Record your answer as a three-digit number.)

$$\sqrt{144}, \quad \pi, \quad -\frac{8}{3}, \quad \sqrt[3]{-125}, \quad 0.\overline{72}, \quad \sqrt{20}, \quad \sqrt{0.01}, \quad 2\sqrt{5}, \quad 0, \quad \sqrt[3]{9}, \quad \frac{\sqrt{75}}{\sqrt{3}}, \quad -\sqrt{1.44}$$

Lesson 6 — Extra Practice

Roots of Real Numbers

Part A — Exact Values Without a Calculator

1. Evaluate exactly. Each radicand is a perfect power of the given index, so the root of the numerator and the root of the denominator can be taken separately. Leave each answer as a fraction in lowest terms.

(a) $\sqrt[3]{\frac{27}{64}}$ (b) $\sqrt[4]{\frac{16}{81}}$ (c) $\sqrt[5]{-\frac{32}{243}}$ (d) $\sqrt[3]{\frac{125}{8}}$

2. Evaluate exactly. Rewrite each decimal as a fraction over a power of ten first — the number of decimal places in the answer is the number of places in the radicand divided by the index.

(a) $\sqrt[4]{0.0016}$ (b) $\sqrt[3]{0.008}$ (c) $\sqrt{0.0064}$ (d) $\sqrt[3]{-0.216}$

3. Evaluate exactly. Watch the coefficient in front of the radical — it multiplies the root *after* the root has been taken.

(a) $\sqrt[6]{729}$ (b) $-2\sqrt[5]{-1\,024}$ (c) $\frac{1}{2}\sqrt[7]{128}$ (d) $\sqrt[6]{\frac{1}{64}}$

4. Evaluate exactly. In each case the radicand is already written as a power, so decide what the power is *first*, then take the root. Two of these do not give back the number inside the bracket.

(a) $\sqrt{(-5)^2}$ (b) $\sqrt[3]{(-5)^3}$ (c) $\sqrt[4]{(-3)^4}$ (d) $-\sqrt{(-7)^2}$

Part B — Possible or Not Possible

5. Give the exact value, or write **"not possible"** if the radical is not a real number. Remember: **if the index is even, the radicand must be non-negative.**

(a) $\sqrt[4]{-16}$ (b) $\sqrt[3]{-216}$ (c) $\sqrt[6]{-64}$ (d) $\sqrt[5]{-243}$

(e) $-\sqrt{49}$ (f) $\sqrt{-49}$ (g) $\sqrt[8]{256}$ (h) $\sqrt[3]{-\frac{1}{1\,000}}$

6. Each pair below looks almost the same, but the minus sign has moved. Evaluate **both** expressions in each pair, then say in one sentence whether the two are equal and why the position of the minus sign matters.

(a) $-\sqrt[4]{81}$ and $\sqrt[4]{-81}$ (b) $\sqrt{(-6)^2}$ and $\left(\sqrt{-6}\right)^2$ (c) $-\sqrt[3]{27}$ and $\sqrt[3]{-27}$

7. Consider the radical $\sqrt[n]{-64}$, where $n \in N$.

(a) Complete the table for $n = 2$ through $n = 7$. Write the exact value when it is an integer, the word *irrational* when the root exists but is not an integer, and **not possible** when it is not a real number.

$n = 2$: _____

$n = 3$: _____

$n = 4$: _____

$n = 5$: _____

$n = 6$: _____

$n = 7$: _____

(b) **Explain** the pattern you see in terms of the index alone, not in terms of 64.

(c) A classmate says "-64 has no roots at all." Correct the statement using your table.

(d) For how many of the six values of n is the answer an *integer*?

Part C — Estimating Irrational Roots

8. Each root below is irrational. **Without a calculator**, state the two consecutive integers it lies between, and write down the two perfect powers you used to bracket it.

(a) $\sqrt{57}$ (b) $\sqrt[3]{100}$ (c) $\sqrt[4]{300}$ (d) $\sqrt[3]{-30}$

9. Now refine each estimate to **one decimal place**, still without a calculator. Test the halfway value first: raise it to the power of the index and compare with the radicand — that tells you at once which way the root rounds. Show the powers you tested.

(a) $\sqrt{57}$ (b) $\sqrt[3]{100}$ (c) $\sqrt{40}$ (d) $\sqrt[3]{-30}$

10. Ivy claims that $\sqrt[3]{80} > \sqrt{15}$ "because 80 is bigger than 15."

(a) Bracket each root between consecutive integers and decide whether her *conclusion* is correct.

(b) Her *reasoning* is still wrong. Give a counterexample: two radicals with different indices where the larger radicand gives the smaller value.

(c) **Explain** why radicands may only be compared directly when the two indices match.

Part D — Ordering Roots

11. Arrange from **least to greatest** without a calculator. Every value here is either an integer or sits between two integers you can name.

$$\sqrt[3]{27}, \qquad \sqrt{15}, \qquad \sqrt[4]{16}, \qquad \sqrt[3]{-8}, \qquad \sqrt{10}$$

12. Arrange from **least to greatest** without a calculator. Simplify every radical first — five of the six are exact — and be careful with the two negatives.

$$\sqrt[3]{-64}, \qquad -\sqrt[4]{81}, \qquad \sqrt{0.25}, \qquad \sqrt[3]{0.064}, \qquad \sqrt[6]{64}, \qquad \sqrt{5}$$

Part E — Error Analysis

13. A student writes: "$\sqrt{-16} = -4$, because $(-4) \times (-4)$ has a negative in it, so it must give -16."

(a) Evaluate $(-4)^2$ and use it to identify the mistake in one sentence.

(b) State the correct answer for $\sqrt{-16}$, using the booklet's wording.

(c) The student's answer of -4 *is* the correct value of a closely related radical. Write that radical.

14. A second student writes: "$\sqrt[3]{-64}$ is not possible, because you can never take the root of a negative number."

(a) Evaluate $\sqrt[3]{-64}$ and verify your answer by cubing it.

(b) State the rule the student should have used, naming the index.

(c) Write one radical with a negative radicand that really *is* not possible, and say why.

Part F — Reasoning

15. Counting roots.

(a) **Explain** why a positive number has *two* square roots but only *one* cube root. Base your explanation on what happens to the sign when a number is multiplied by itself twice versus three times.

(b) 64 has two square roots, yet $\sqrt{64} = 8$ only. **Explain** what the radical sign $\sqrt{\ }$ is defined to mean, and why a single symbol cannot stand for two numbers at once.

(c) How many real fourth roots does 81 have? How many real fifth roots does 32 have? State the general rule for an even index and for an odd index.

16. The identity $\sqrt{x^2} = |x|$.

(a) Complete the table.

x	x^2	$\sqrt{x^2}$
5	_____	_____
0	_____	_____
-5	_____	_____
$-\frac{3}{4}$	_____	_____

(b) **Explain** why $\sqrt{x^2} = |x|$ rather than $\sqrt{x^2} = x$. Your explanation should say what squaring does to a negative number and what the principal root then gives back.

(c) For which real values of x is $\sqrt{x^2} = x$ actually true?

(d) Is $\sqrt[3]{x^3} = x$ true for *every* real x? Test $x = -5$ and explain the difference.

17. **Stretch.** Solve each equation over the real numbers.
(a) $x^3 = -125$
(b) $x^4 = 81$
(c) $x^4 = -81$
(d) $x^5 = 32$

(e) **Explain** why (a) and (b) have different *numbers* of solutions even though both radicands are perfect powers, and why (c) has none at all. Then state, in one sentence, how many real solutions $x^n = k$ has for an even n and for an odd n.

18. **Stretch — nested radicals.** Work from the inside out.
(a) $\sqrt{\sqrt{81}}$ (b) $\sqrt[3]{\sqrt{64}}$ (c) $\sqrt{\sqrt[3]{729}}$

(d) Rewrite $\sqrt{\sqrt[3]{x}}$ as a *single* radical, and state the rule that lets you combine the two indices. Check your rule against part (a).

Part G — Multiple Choice and Numerical Response

19. *(Multiple Choice)* Which of the following is **not possible** in the real number system?
A. $\sqrt[3]{-\frac{8}{27}}$ **B.** $\sqrt[4]{-\frac{16}{81}}$
C. $-\sqrt[4]{\frac{16}{81}}$ **D.** $\sqrt[5]{-\frac{1}{32}}$

20. *(Multiple Choice)* Consider these four statements.
i) $\sqrt{(-8)^2} = -8$
ii) $\sqrt[3]{(-8)^3} = -8$
iii) $-\sqrt[6]{64} = \sqrt[6]{-64}$
iv) $\sqrt[5]{-1} = -1$
Which are **true**?
A. ii) and iv) only **B.** i), ii) and iv) only
C. ii), iii) and iv) only **D.** all four

21. *(Numerical Response)* The value of
$$\sqrt[4]{50\,625} + \sqrt[3]{-1\,728} + \sqrt{(-9)^2}$$
is _____. *(Record your answer as a three-digit number.)*

Lesson 7 — Extra Practice

Entire and Mixed Radicals: From Square Roots to Higher-Index Roots

Part A — Large Radicands and Both Directions

1. Convert each to a mixed radical in simplest form. These radicands are large enough that guessing a perfect square is slow — write the prime factorization first, then pair the primes off.

(a) $\sqrt{720}$ (b) $\sqrt{1\,008}$ (c) $\sqrt{2\,700}$ (d) $\sqrt{4\,056}$

2. Convert each to a mixed radical in simplest form. Simplify the radical first, then deal with the coefficient. One of these turns out not to be a mixed radical at all — say which, and why.

(a) $-2\sqrt{50}$ (b) $\frac{3}{8}\sqrt{1\,152}$ (c) $-\frac{5}{6}\sqrt{4\,356}$ (d) $\frac{5}{12}\sqrt{6\,912}$

3. Here the radicand is *already* prime-factored, which is the whole point — an even exponent comes straight out, an odd exponent leaves one factor behind. Write each as a mixed radical in simplest form.

(a) $\sqrt{2^5 \times 3^4 \times 5^2}$ (b) $\sqrt{3^5 \times 7^2 \times 11}$ (c) $\sqrt[3]{2^7 \times 3^4 \times 5}$

4. Convert each mixed radical to an entire radical.

(a) $7\sqrt{10}$ (b) $-8\sqrt{5}$ (c) $\frac{3}{4}\sqrt{96}$ (d) $\frac{2}{5}\sqrt{150}$ (e) $\frac{5}{2}\sqrt{20}$ (f) $0.4\sqrt{75}$

5. Three of the four radicals below can be written as mixed radicals; one cannot. Simplify the three, then **explain** what is true about the prime factorization of the fourth radicand.

(a) $\sqrt{1\,183}$ (b) $\sqrt{2\,450}$ (c) $\sqrt{1\,590}$ (d) $\sqrt{5\,292}$

Part B — Index 3 and Higher

6. **The index changes what you pull out.** For a cube root you need the largest perfect *cube* factor, not the largest perfect square. Convert each to a mixed radical in simplest form.

(a) $\sqrt[3]{54}$ (b) $\sqrt[3]{1\,512}$ (c) $\sqrt[3]{-128}$ (d) $\sqrt[3]{-3\,456}$

7. Same idea at index 4 and index 5: group the prime factors into fours or fives.
(a) $\sqrt[4]{2\,592}$ (b) $\sqrt[4]{6\,912}$ (c) $\sqrt[5]{3\,888}$ (d) $\sqrt[5]{-2\,048}$

8. Convert to a mixed radical in simplest form. Watch the sign travelling out of an odd root.
(a) $-4\sqrt[3]{5\,184}$ (b) $\frac{3}{5}\sqrt[4]{1\,250}$ (c) $2\sqrt[5]{-2\,048}$ (d) $\frac{5}{7}\sqrt[3]{6\,174}$

9. Convert each mixed radical to an entire radical. The coefficient is raised to the power of the *index* — and with an even index a negative coefficient has to stay outside.
(a) $5\sqrt[3]{2}$ (b) $-6\sqrt[3]{4}$ (c) $\frac{3}{2}\sqrt[3]{16}$ (d) $2\sqrt[4]{6}$ (e) $-5\sqrt[4]{3}$ (f) $\frac{1}{2}\sqrt[5]{96}$

10. Evaluate each exactly, or state that it is **not possible** in the real number system. Where it is not possible, say which feature of the radical makes it so.
(a) $\sqrt[3]{-1\,000}$ (b) $\sqrt[4]{-16}$ (c) $-\sqrt[4]{16}$ (d) $\sqrt[5]{-243}$

Part C — Comparing Radicals

11. **(No calculator.)** Arrange from **least to greatest**. Convert every radical to entire form first, then compare the radicands. Show the entire form you used for each.

$$3\sqrt{7}, \qquad 2\sqrt{15}, \qquad 5\sqrt{2}, \qquad \frac{1}{2}\sqrt{180}, \qquad \sqrt{71}$$

12. **(No calculator.)** Arrange from **least to greatest**. Every radical here has index 3, so the same strategy works — but the coefficient is now *cubed*, not squared.

$$2\sqrt[3]{5}, \qquad \sqrt[3]{50}, \qquad 3\sqrt[3]{2}, \qquad \frac{1}{2}\sqrt[3]{448}, \qquad \frac{1}{3}\sqrt[3]{1\,539}$$

Part D — Error Analysis

13. Each student below made exactly one mistake. For each, **name the error** and give the correct answer.

(a) **Nadia** simplifies a cube root:
$\sqrt[3]{72} = \sqrt[3]{36 \times 2} = 6\sqrt[3]{2}$

(b) **Owen** writes $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$, then says "and $6\sqrt{2}$ simplifies further to $\sqrt{12}$."

(c) **Priya** converts to an entire radical:
$-3\sqrt[4]{5} = \sqrt[4]{-405}$

(d) **Quinn** writes $\sqrt{720} = \sqrt{16 \times 45} = 4\sqrt{45}$ and stops there.

14. *(Multiple Choice)* Three of the four expressions below are equal to $\sqrt[3]{432}$, but only one of them is written as a mixed radical **in simplest form**. Which one? A. $2\sqrt[3]{54}$ B. $3\sqrt[3]{16}$ C. $6\sqrt[3]{2}$ D. $4\sqrt[3]{7}$

Part E — Radicals With Variables

Assume every variable represents a non-negative number.

15. Express each as a mixed radical in simplest form. Divide each exponent by the index: the quotient comes out, the remainder stays under.

(a) $\sqrt{288a^9b^6}$ (b) $\sqrt[3]{432x^{11}y^7}$ (c) $-2\sqrt{700c^5d^9}$ (d) $\sqrt[4]{1\,250m^{13}n^4}$

16. Express each as an entire radical.

(a) $3x^2\sqrt{2x^3}$ (b) $5ab^3\sqrt{3a^2b}$ (c) $2pq^2\sqrt[3]{5pr^4}$ (d) $\frac{2}{3}m^3\sqrt{18mn^2}$

Part F — Combining, Applying, Stretching

17. **Beyond the booklet: like radicals.** Two radicals are **like radicals** when they share the *same index and the same radicand*, and only then can their coefficients be added or subtracted: $3\sqrt{5}+4\sqrt{5}=7\sqrt{5}$, exactly as $3x+4x=7x$. The radicals below look unlike, so **simplify each term to a mixed radical first**, then combine.

(a) $\sqrt{50}+\sqrt{18}$ (b) $3\sqrt{12}-\sqrt{75}$ (c) $\sqrt{45}+\sqrt{80}-\sqrt{20}$ (d) $2\sqrt[3]{54}+\sqrt[3]{16}$

18. Exact radical answers in geometry.

(a) A rectangular box measures 6 cm by 8 cm by 12 cm. The longest straight rod that fits inside runs corner to corner, and its length d satisfies $d^2=6^2+8^2+12^2$.
(i) Give d as an entire radical.
(ii) Give d as a mixed radical in simplest form.
(iii) Give d to the nearest hundredth of a centimetre.

(b) An equilateral triangle has side length 12 cm. Dropping a perpendicular from one vertex splits it into two right triangles.
(i) Find the exact height as a mixed radical.
(ii) Find the exact area as a mixed radical.

19. *(Numerical Response)* Answer each in the blank beside it.

(a) Written in simplest form, $\sqrt{3\,528}=a\sqrt{b}$ where $a,b\in N$. The value of $a+b$ is _____.

(b) The box diagonal from question 18(a) is $p\sqrt{q}$ in simplest form, where $p,q\in N$. The value of pq is _____.

20. **Stretch.** These ask you to run the whole process backwards.

(a) Find the *smallest* whole number n for which $\sqrt{n}$, in simplest form, is a mixed radical with coefficient exactly 6. Justify why nothing smaller works.

(b) Simplify $\sqrt{10!}$, where $10!=10\times 9\times 8\times\cdots\times 2\times 1$. Do **not** multiply it out — factor each of 2 through 10 into primes and count.

(c) Find the smallest whole number n for which $\sqrt[3]{n}$, in simplest form, is a mixed radical with coefficient exactly 5. Explain how the index changed your search.

Lesson 9 — Extra Practice

Number Unit Review

Part A — Factorization and Number Theory

1. Let $m = 2^6 \times 3^4 \times 5^2$ and $n = 2^3 \times 3^4 \times 7$. Answer every part **from the exponents** — do not multiply out unless a part asks you to.

(a) How many factors does m have? How many does n have?

(b) Is m a perfect square? Is it a perfect cube? Justify each answer using the exponents only.

(c) Determine the GCF and the LCM of m and n. Give each in exponent form *and* as a single number.

(d) Is $\sqrt{m}$ rational? If it is, give its exact value; if it is not, write it in simplest mixed radical form.

(e) Is $\sqrt{n}$ rational? Write it in simplest form, then list *every* set from $N, W, I, Q, \overline{Q}, R$ that it belongs to.

(f) Is $\sqrt[3]{m}$ rational? Write it in simplest mixed radical form and classify it.

2. Let $p = 2^5 \times 3^2 \times 7^3$.

(a) Write p as a single number, and state how many factors it has.

(b) Find the **smallest** natural number k for which pk is a perfect square, and state $\sqrt{pk}$.

(c) Find the **smallest** natural number k for which pk is a perfect cube, and state $\sqrt[3]{pk}$.

(d) Write $\sqrt{p}$ in simplest mixed radical form, and list every set it belongs to.

(e) **Explain** how both answers in (b) and (c) can be read straight off the exponents of p, without any guessing and without a calculator.

3. Let $a = 2^3 \times 3 \times 5^2$ and $b = 2^2 \times 3^3 \times 5$.

(a) Determine $\text{GCF}(a, b)$ and $\text{LCM}(a, b)$ in exponent form.

(b) Verify that $\text{GCF}(a, b) \times \text{LCM}(a, b) = ab$ by evaluating both sides.

(c) Reduce $\frac{a}{b}$ to lowest terms. Which part of (a) did you just use?

(d) Factor the *reduced* denominator and use it to predict whether $\frac{a}{b}$ terminates or repeats. Then write out the decimal and check your prediction.

(e) List every set from $N, W, I, Q, \overline{Q}, R$ that $\frac{a}{b}$ belongs to, and name the single *strictest* one.

4. Let $N = 2\,646$.

(a) Find the prime factorization of N in exponent form, and state how many factors N has.

(b) Is N a perfect square? A perfect cube? Answer from the exponents.

(c) Write $\sqrt{N}$ in simplest mixed radical form, and classify it.

(d) Find the smallest natural number k for which kN is a perfect square, and state $\sqrt{kN}$.

(e) **Explain** the connection between your answer to (c) and your answer to (d): what does the radicand left under the root sign in (c) tell you about k?

Part B — Decimals and Classification

5. Take each fraction through the whole chain. For each one: **(i)** reduce it to lowest terms, **(ii)** factor the *reduced* denominator, **(iii)** predict terminating or repeating from that factorization, **(iv)** write the decimal (use bar notation if it repeats), and **(v)** list every set it belongs to.

(a) $\frac{126}{360}$

(b) $\frac{84}{385}$

(c) $\dfrac{1\,044}{174}$ (d) $\dfrac{1\,183}{1\,001}$

6. Now run the chain *backwards*. For each decimal: convert it to a fraction in lowest terms, factor the denominator to confirm that the decimal *had* to repeat, and list every set the number belongs to.
(a) $0.\overline{36}$ (b) $2.1\overline{6}$ (c) $0.\overline{123}$ (d) $0.\overline{9}$

(e) Part (d) breaks the pattern of the other three. **Explain** what happens to the denominator test there, and state every set $0.\overline{9}$ belongs to.

7. Simplify each number first, then sort the eight of them into three groups: **rational**, **irrational**, and **not possible in the real number system**.

$$\sqrt{0.16}, \quad \sqrt[3]{-\frac{8}{27}}, \quad \sqrt{40}, \quad 0.1\overline{6}, \quad \sqrt[4]{-16}, \quad \frac{\sqrt{98}}{\sqrt{2}}, \quad \frac{\pi}{\pi}, \quad 2.010010001\ldots$$

Part C — Radicals

8. Simplify each radical completely. Then state whether the result is **rational**, and list every set from $N, W, I, Q, \overline{Q}, R$ that it belongs to.
(a) $\sqrt{392}$ (b) $\sqrt[3]{-432}$ (c) $\sqrt[4]{405}$

(d) $\sqrt{\dfrac{242}{2}}$

(e) $\sqrt[3]{\dfrac{-54}{2}}$

(f) $\sqrt[4]{-81}$

9. Conversions and comparisons. Keep every answer exact.

(a) Write $7\sqrt{6}$ as an entire radical.

(b) Write $2\sqrt[3]{9}$ as an entire radical.

(c) Write $-3\sqrt[4]{2}$ as an entire radical. Where does the negative sign have to stay, and why?

(d) Which is larger, $5\sqrt{3}$ or $3\sqrt{7}$? Decide *without* a calculator.

(e) Which is larger, $2\sqrt[3]{10}$ or $\sqrt[3]{81}$? Again, no calculator.

(f) Is $\sqrt{\sqrt{81}}$ rational? Simplify it and name the strictest set it belongs to.

10. A square has area 180 cm^2 and a cube has volume 1 029 cm^3.

(a) Find the exact side length of the square in simplest mixed radical form, then classify it.

(b) Find the exact edge length of the cube in simplest mixed radical form, then classify it.

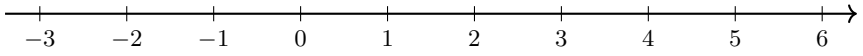
(c) Find the exact perimeter of the square. Is a rational multiple of an irrational number ever rational? Use your answer to explain.

11. Consider the eight numbers

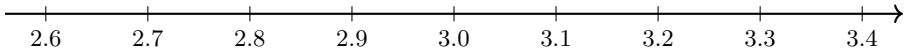
$$\frac{13}{4},\qquad -\sqrt{5},\qquad 2.\overline{7},\qquad \sqrt[3]{30},\qquad -2,\qquad \frac{\sqrt{72}}{\sqrt{2}},\qquad \sqrt{10},\qquad -\frac{7}{3}.$$

(a) Write each as a decimal correct to three decimal places, then list all eight in order from **least to greatest**.

(b) Plot all eight on the number line below, labelling each point with the number in its *original* form.



(c) Three of them land between 3 and 3.3, too close together to separate above. Plot just those three on this magnified line.



(d) Which of the eight numbers are in $\overline{Q}$? How can you tell without a calculator?

Part D — Reasoning and Error Analysis

12. State whether each claim is **true** or **false**. If it is true, give the reason it must always hold; if it is false, give a counterexample.

(a) Every non-terminating decimal is irrational.

(b) For a natural number n, $\sqrt{n}$ is irrational unless n is a perfect square.

(c) If every exponent in a natural number's prime factorization is even, its square root is a natural number.

(d) The product of two irrational numbers is irrational.

(e) A radical with an odd index always represents a real number.

(f) An entire radical is always irrational.

(g) If $\text{GCF}(a, b) = 1$, then $\text{LCM}(a, b) = ab$.

(h) If a fraction's denominator contains a prime other than 2 or 5, its decimal repeats.

13. A student was asked to simplify $\sqrt{180}$ and classify the result. Here is the entire submission.

$$\sqrt{180} = \sqrt{9 \times 20} = \sqrt{9} \times \sqrt{20} = 3\sqrt{20}$$

"Because it still has a radical sign, it is irrational, so it belongs to $\overline{Q}$ and that is the only set it is in."

(a) There are *three* separate errors here. Identify each one in a single sentence.

(b) Give the correct simplest mixed radical form, showing the choice of perfect square that ends the job in one step.

(c) List every set the simplified number really belongs to, and state the reliable test for deciding whether $\sqrt{n}$ is irrational.

14. A second student is given $a = 2^3 \times 5$ and $b = 2 \times 5^2$ and writes:
"$\text{GCF}(a, b) = 2^3 \times 5^2 = 200$ and $\text{LCM}(a, b) = 2 \times 5 = 10$. So $\frac{a}{b} = \frac{40}{50}$, and since the denominator $50 = 2 \times 5^2$ contains a 2 and a 5, the decimal repeats."

(a) The GCF and LCM have been swapped. State the correct values and the rule that decides which exponent goes where.

(b) The denominator test has been misremembered *and* misapplied. State the rule correctly, and name the step the student skipped before applying it.

(c) Give the correct decimal for $\frac{a}{b}$ and list every set it belongs to.

15. Statement 12(b) is worth arguing properly. Let n be a natural number.

(a) **Explain**, using prime factorizations and exponents alone, why $\sqrt{n}$ is a natural number exactly when every exponent in n is even — and why $\sqrt{n}$ can therefore never be a fraction like $\frac{7}{2}$ that is rational but not whole.

(b) Test your reasoning on $n = 2^4 \times 3^3 \times 5^2$: name the exponent that spoils it, write $\sqrt{n}$ in simplest mixed radical form, and classify the result.

Part E — Challenge

16. A natural number n is chosen so that $\sqrt{n}$, written in simplest mixed radical form, has coefficient exactly 12.

(a) Find the **smallest** such n, and write $\sqrt{n}$ in simplest mixed radical form.

(b) A classmate suggests $n = 144 \times 4 = 576$. **Explain** why that fails, and state the condition the leftover radicand must satisfy.

(c) Find the next *two* values of n after your answer to (a), in increasing order.

(d) Is $\sqrt{n}$ rational for your answer in (a)? Name every set it belongs to.

17. Two natural numbers a and b with $a < b$ satisfy $\text{GCF}(a, b) = 12$ and $\text{LCM}(a, b) = 180$.

(a) Use $\text{GCF} \times \text{LCM} = ab$ to find ab, then write $a = 12m$ and $b = 12n$ and explain what m and n must satisfy. Find *every* possible pair (a, b).

(b) For each pair you found, reduce $\frac{a}{b}$ to lowest terms, predict terminating or repeating from the reduced denominator, write the decimal, and list every set the quotient belongs to.

18. *(Multiple Choice)* A number $k = 2^a \times 3^b$ is a perfect square and has exactly 15 factors. The smallest possible value of k is

A. 36 **B.** 144 **C.** 324 **D.** 1 296

19. *(MC)* Written in simplest mixed radical form, $\sqrt[3]{-1\,080}$ is

A. $-6\sqrt[3]{5}$ **B.** $-2\sqrt[3]{135}$

C. $-3\sqrt[3]{40}$ **D.** not possible in the real number system

20. *(Numerical Response)* Record each answer as a three-digit number. Part (b) refers to the twelve numbers listed here.

$$\sqrt{162},\quad \sqrt[3]{-64},\quad 0.\overline{81},\quad \frac{\sqrt{200}}{\sqrt{2}},\quad \sqrt[4]{625},\quad 2\sqrt{9},\quad \pi + 1,\quad \sqrt{\frac{49}{100}},\quad \sqrt[3]{20},\quad \frac{22}{7},\quad \sqrt{1.21},\quad 3.14114111\ldots$$

(a) For $x = 2^5 \times 3^4 \times 7^2$, the smallest natural number k such that kx is a perfect cube is _____.

(b) The number of the twelve numbers above that are irrational is _____.